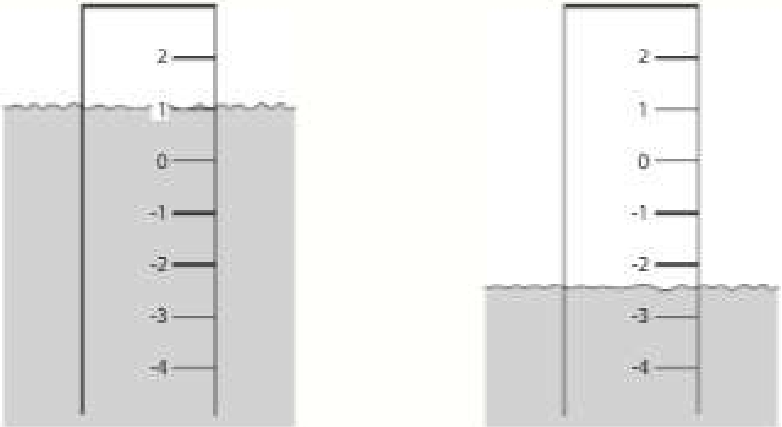
2 The diagram shows how the water level of a river went down during a drought.



The measurements are in metres.

(a) By how many metres did the water level go down?

Level goes from 1 to -2.5 metres

$1 - (-2.5) = 3.5$

Answer (a) 3.5 m [1]

(b) A heavy rainfall followed the drought and the water level went up by 1.6 metres.
What was the water level after the rainfall?

$-2.5 + 1.6$

Answer (b) -0.9 m [1]

---

3 **(a)** Write in order of size, smallest first

$0.68 = \frac{68}{100}$ 0.68, $\frac{33}{50}$, 67%. $67\% = \frac{67}{100}$

$\frac{33}{50} = \frac{66}{100}$

Answer (a) $\frac{33}{50}$ < 67% < 0.68 [1]

(b) Convert 0.68 into a fraction in its lowest terms.

$\frac{68}{100} = \frac{34}{50} = \frac{17}{25}$

Answer (b) $\frac{17}{25}$ [1]

For Examiner's Use

4 Mahesh and Jayraj share \$72 in the ratio 7:5.
How much does Mahesh receive?

\$72 in the ratio 7 : 5
7 + 5 = 12 parts
1 part = 72 ÷ 12 = 6
Mahesh receives 7 parts which is 7 × 6

Answer \$ 42 .... [2]

---

5 The population of a city is 550 000.
It is expected that this population will increase by 42% by the year 2008.
Calculate the expected population in 2008.

42% of 550 000
= 42 ÷ 100 × 550 000
= 231 000
Expected population in 2008
= 550 000 + 231 000
= 781 000

Answer 781 000 .... [2]

---

6 Areeg goes to a bank to change \$100 into riyals.
The bank takes \$2.40 and then changes the rest of the money at a rate of \$1 = 3.75 riyals.
How much does Areeg receive in riyals?

\$100 − \$2.40 = \$97.60
Since \$1 = 3.75 Riyals
\$97.60 = 3.75 × 97.60 = 366

Answer 366 ....riyals [2]

---

7 Write down the value of $\left(1\frac{1}{2}\right)^{-2}$ as a fraction.

$\left(1\frac{1}{2}\right)^{-2} = 1/\left(1\frac{1}{2}\right)^2$
$= 1/\left(\frac{3}{2}\right)^2$
$= 1/\left(\frac{9}{4}\right)$ $= \frac{4}{9}$

Answer $\frac{4}{9}$ .... [2]

---

8 **(a)** $y = 4uv - 3v$.

Find the value of y when $u = -3$ and $v = 2$.

$y = 4uv - 3v$
$= 4 \times (-3) \times 2 - 3 \times 2$
$= -24 - 6$ $= -30$

Answer (a) $y =$ −30 .... [1]

(b) Factorise $4uv - 3v$.

$4uv - 3v = v(4u - 3)$

Answer (b) $v(4u - 3)$ .... [1]

For Examiner's Use

$$x + 4 = 3(2 - x).$$

$x + 4 = 3(2 - x)$

$x + 4 = 6 - 3x$

$x + 3x = 6 - 4$

$4x = 2$

$x = \frac{1}{2}$ or 0.5

Answer x = $\frac{1}{2}$ or 0.5 .... [3]

---

10 There are approximately 500 000 grains of wheat in a 2 kilogram bag.

(a) Calculate the mass of one grain in grams.

2kg = 2000g

2000 ÷ 500 000

= 0.004

Answer (a) 0.004 ....g [2]

(b) Write your answer to **part (a)** in standard form.

$0.004 = 4 \times 10^{-3}$

Answer (b) 4×10^{-3} ....g [1]

---

11 Solve the simultaneous equations $3a + 2b = 7$,
$a - 2b = 5$.

$3a + 2b = 7$

$a - 2b = 5$

Adding the equations,

$3a + a + 2b - 2b = 7 + 5$

$4a = 12$

$a = 12/4$

$a = 3$

Substituting $a = 3$ in $3a + 2b = 7$,

$3 \times 3 + 2b = 7$

$9 + 2b = 7$

$2b = -2$

$b = -1$

Answer a = 3 ....

b = -1 .... [3]

For Examiner's Use

12 The diagram shows a pole of length l centimetres.

(a) Hassan says that $l = 88.2$.
Round this to the nearest whole number.

Answer (a) l = 88 [1]

(b) In fact the pole has a length 86 cm, to the nearest centimetre.
Complete the statement about l.

Answer (b) 85.5 $\leqslant l <$ 86.5 [2]

---

13 On a journey a bus takes 35 minutes to travel the first 10 kilometres.
It then travels a further 20 kilometres in the next 40 minutes.

(a) The bus started the journey at 18 50.
At what time did it complete the journey?

The journey ends 75 minutes later.
75 minutes is 1 hour 15 minutes
1 hour 15 minutes after 18 50 is 20 05

Answer (a) 20 05 [1]

(b) Calculate the average speed of the whole journey in

(i) kilometres/minute,

Average speed = (30 ÷ 75) km per minute
= 0.4

Answer (b)(i) 0.4 km/min [2]

(ii) kilometres/hour.

0.4 × 60 = 24

Answer (b)(ii) 24 km/h [1]

For Examiner's Use

14 Show **all your working** for the following calculations.
The answers are given so it is only your working that will be given marks.

(a) $\frac{1}{2} + \frac{2}{3} = 1\frac{1}{6}$.

Answer (a) $\frac{1}{2} = \frac{3}{6}$ and $\frac{2}{3} = \frac{4}{6}$

$\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$

$\frac{7}{6} = 1\frac{1}{6}$ [2]

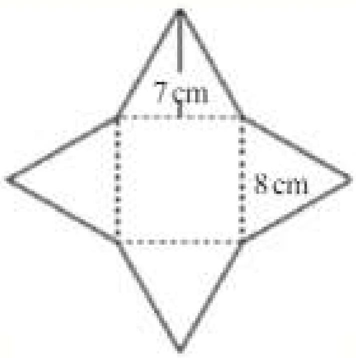
(b) $1\frac{1}{5} \times 1\frac{3}{4} = 2\frac{1}{10}$.

Answer (b) $1\frac{1}{5} = \frac{6}{5}$ and $1\frac{3}{4} = \frac{7}{4}$

$\frac{6}{5} \times \frac{7}{4} = (6 \times 7)/(5 \times 4)$ $= \frac{42}{20} = 2\frac{2}{20} = 2\frac{1}{10}$

[2]

15 The diagram shows a square of side 8 cm and four congruent triangles of height 7 cm.



(a) Calculate

(i) the area of one triangle,

Area $= \frac{1}{2}$(base $\times$ height)
$= 0.5 \times 8 \times 7$ $= 28$

Answer (a)(i) ....28.......... cm^2 [2]

(ii) the area of the whole shape.

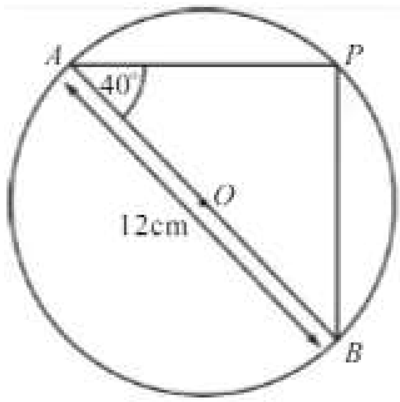
Area of the square $= 8 \times 8$
Whole area $= 4 \times 28 + 8 \times 8$
$= 176$

Answer (a)(ii) ....176.......... cm^2 [2]

(b) The shape is the net of a solid.
Write down the special name for this solid.

Answer (b) ....Pyramid.......... [1]

16 In the diagram AB is the diameter of a circle, centre O. The length of AB is 12cm.



NOT TO SCALE

(a) Write down the size of angle APB.

Answer (a) Angle APB = ...90°.......... [1]

(b) Angle $PAB = 40°$.
Calculate the length of PB.

Sin 40° = PB x 12
PB = 12sin 40°
PB = 7.71

Answer (b) PB = ...7.71..........cm [2]

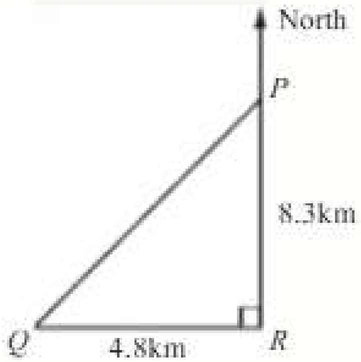
(c) Calculate the area of the circle.

Radius = 6cm
Area = π x (radius)^2
= π x 6^2 = 113

Answer (c) ....113..........cm^2 [2]

For Examiner's Use

17



NOT TO SCALE

A straight road between P and Q is shown in the diagram.
R is the point south of P and east of Q.
$PR = 8.3$ km and $QR = 4.8$ km.

Calculate

(a) the length of the road PQ,

$PQ^2 = 4.8^2 + 8.3^2$
$= 91.93$
$PQ = \sqrt{91.93}$
$= 9.59$

Answer (a) ....9.59............km [2]

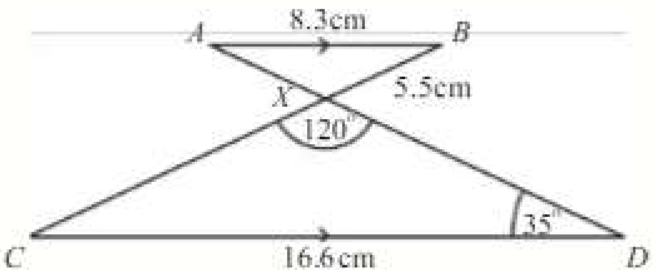
(b) the bearing of Q from P.

$\tan QPR = 4.8 \div 8.3$
$= 0.5783132253$
Angle $QPR = 30.0°$
Bearing of Q from P $= 180 + 30$
$= 210°$

Answer (b) ....210°............ [3]

For Examiner's Use

18



In the diagram the lines AB and CD are parallel.
The lines AD and BC intersect at X.
Angle $XDC = 35°$ and angle $CXD = 120°$.

(a) **(i)** Write down the size of angle BAX.

Answer(a)(i) Angle BAX = ...35°........... [1]

(ii) Write down the size of angle ABX.

Answer(a)(ii) Angle ABX = ...25°........... [1]

(b) Complete the statement

Triangle AXB is ....Similar........... to triangle DXC. [1]

(c) $AB = 8.3$cm, $BX = 5.5$cm and $CD = 16.6$cm.
Calculate the length of CX.

For similar triangles CD/AB = CX/BX
16.6/8.3 = CX/5.5
CX = (16.6 x 5.5)/8.3
CX = 11

Answer (c) ....11........... cm [2]